National University of Computer and Emerging Sciences

Remaining Useful Life (RUL) Prediction for Turbofan Engines with Uncertainty Quantification

F26-16

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Problem Statement

Aircraft turbofan engines are complex mechanical systems that degrade progressively over their operational life. Estimating the Remaining Useful Life (RUL), the number of operating cycles left before failure, is a central task in Prognostics and Health Management (PHM), enabling condition-based maintenance that avoids both unscheduled in-service failures and unnecessarily early part replacement.

Data-driven RUL models trained on multivariate sensor time-series collected across an engine's run-to-failure trajectory have achieved strong accuracy, with deep learning architectures such as LSTMs and 1D-CNNs reporting low RMSE on the NASA C-MAPSS benchmark. However, the great majority of these models output only a single point estimate of RUL and provide no indication of how reliable that estimate is. In a maintenance setting, a point prediction that silently overestimates remaining life can lead directly to an in-service failure, while a model that cannot express when it is uncertain offers limited value for risk-informed maintenance scheduling.

This project therefore investigates Uncertainty Quantification (UQ) for RUL prediction: in addition to predicting a point value for RUL, the model should also output a calibrated confidence interval reflecting how much that prediction should be trusted. The goal is to evaluate whether established UQ techniques, specifically Monte Carlo Dropout and Deep Ensembles, can be layered on top of standard RUL prediction architectures to produce well-calibrated, informative uncertainty estimates without compromising point-prediction accuracy.

Dataset

The project uses the NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) Turbofan Engine Degradation Simulation Data Set, provided by the NASA Prognostics Center of Excellence (PCoE). It contains simulated multivariate sensor readings (21 sensors plus 3 operational settings, recorded per operating cycle) for a fleet of turbofan engines, each run from a healthy state to failure. The data set is split into four sub-datasets that vary in the number of operating conditions and fault modes:

Sub-dataset	Operating Conditions	Fault Modes	Train Engines	Test Engines
FD001	1 (Sea Level)	1 (HPC Degradation)	100	100
FD002	6	1 (HPC Degradation)	260	259
FD003	1 (Sea Level)	2 (HPC + Fan Degradation)	100	100
FD004	6	2 (HPC + Fan Degradation)	248	249

Role in study: primary training and evaluation dataset for the RUL regression and uncertainty-quantification models. Development and initial model selection are carried out on FD001 (the simplest

sub-dataset, single operating condition and single fault mode); the same pipeline is then extended to FD002-FD004 to test generalization under multiple operating conditions and fault modes.

Source: NASA Prognostics Center of Excellence (PCoE) Data Set Repository

Dataset paper: A. Saxena, K. Goebel, D. Simon, N. Eklund (2008), "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation," International Conference on Prognostics and Health Management (PHM08).
https://c3.ndc.nasa.gov/dashlink/static/media/publication/2008_IEEEPHM_CMAPPSDamagePropagation.pdf

Relevant Papers

[1] Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation — Saxena, Goebel, Simon & Eklund (2008)

Describes the C-MAPSS simulation and damage-propagation model used to generate the dataset this project is built on. Essential background for understanding how RUL ground truth is defined and what the sensor channels represent.

https://c3.ndc.nasa.gov/dashlink/static/media/publication/2008_IEEEPHM_CMAPPSDamagePropagation.pdf

[2] Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning — Gal & Ghahramani (2016)

Shows that applying dropout at test time (Monte Carlo Dropout) approximates Bayesian inference in a neural network, giving a simple way to obtain predictive uncertainty. This is one of the two core UQ methods this project implements and compares.

https://arxiv.org/abs/1506.02142

[3] Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles — Lakshminarayanan, Pritzel & Blundell (2017)

Proposes training an ensemble of independently initialized networks as a non-Bayesian alternative for estimating predictive uncertainty. Used as the second UQ method in this project, to be compared against MC Dropout for calibration quality.

https://arxiv.org/abs/1612.01474

[4] Attention and Long Short-Term Memory Network for Remaining Useful Lifetime Predictions of Turbofan Engine Degradation — de Oliveira da Costa, Akcay, Zhang & Kaymak (2019)

Applies an LSTM with an attention mechanism to C-MAPSS RUL prediction and reports competitive results against prior LSTM-based approaches (including Zheng et al., 2017). Informs the architecture choice for this project's deep learning baseline.

https://papers.phmsociety.org/index.php/ijphm/article/view/2623

[5] Uncertainty in Aircraft Turbofan Engine Prognostics on the C-MAPSS Dataset — PHM Society (PHME Conference)

Surveys aleatoric and epistemic uncertainty in data-driven RUL prognostics and compares a stochastic model against an LSTM model directly on C-MAPSS, providing a UQ-focused benchmark closely aligned with this project's objective.

https://papers.phmsociety.org/index.php/phme/article/download/4007/2471

[6] A Parallel Prognostic Method Integrating Uncertainty Quantification for Probabilistic Remaining Useful Life Prediction of Aero-Engine — Processes, MDPI (2024)

Integrates Monte Carlo Dropout into a dual-channel deep learning framework for probabilistic RUL prediction on C-MAPSS, demonstrating a concrete architecture for combining feature extraction with UQ.

https://mdpi-res.com/d_attachment/processes/processes-12-02925/article_deploy/processes-12-02925.pdf

[7] Uncertainty-Aware Deep Learning Framework for Remaining Useful Life Prediction in Turbofan Engines with Learned Aleatoric Uncertainty — arXiv preprint (2025)

Proposes learning aleatoric uncertainty directly via simultaneous mean and variance prediction in a CNN-LSTM hybrid, offering a complementary UQ approach (separate from MC Dropout and ensembles) that this project can discuss or compare against.

https://arxiv.org/abs/2511.19124

[8] Data-driven Prognostics with Predictive Uncertainty Estimation using Ensemble of Deep Ordinal Regression Models — IBM Research

Uses an ensemble of LSTM-based ordinal regression models to obtain predictive uncertainty on C-MAPSS, and shows that uncertainty-aware training can also improve point RUL accuracy, motivating the ensemble-based UQ component of this project.

https://cs.paperswithcode.com/paper/data-driven-prognostics-with-predictive

Project Scope

This project focuses on building a Remaining Useful Life prediction pipeline for turbofan engines on the NASA C-MAPSS dataset, and on augmenting it with uncertainty quantification so that every RUL prediction is accompanied by a calibrated confidence interval. The scope spans data preprocessing, point-prediction modeling, uncertainty-aware modeling, and calibration evaluation; it does not include real-time deployment or hardware-in-the-loop testing.

Models to be Used

- Random Forest / XGBoost: tree-based baseline on windowed, hand-engineered sensor features, used as a reference point-prediction accuracy.

- LSTM (and/or 1D-CNN): deep sequence model trained directly on sliding windows of raw sensor readings, serving as the main point-prediction architecture.
- MC Dropout: dropout layers kept active at inference time; multiple stochastic forward passes over the trained LSTM yield a predictive distribution over RUL.
- Deep Ensemble: several independently initialized and trained LSTM models whose predictions are combined to estimate both the mean RUL and the predictive uncertainty.

Project Overview (Scope of Pipeline)

- 1. Data Preparation: load FD001-FD004, label RUL per cycle (piecewise-linear cap around 125-130 cycles), drop constant/uninformative sensors, and normalize (per operating-condition regime for FD002/FD004).
- 2. Baseline Model Training: train Random Forest / XGBoost on windowed features; evaluate RMSE and the NASA asymmetric PHM08 scoring function.
- 3. Deep Model Training: train an LSTM (and/or 1D-CNN) on sliding-window sensor sequences as the main point-prediction model.
- 4. Uncertainty Quantification: implement MC Dropout and Deep Ensembles on top of the trained deep model to produce prediction intervals.
- 5. Evaluation: assess point-prediction accuracy (RMSE, NASA score) and uncertainty quality (prediction interval coverage probability, average interval width, calibration plots).
- 6. Extension and Analysis: re-run the pipeline on FD002/FD004 (multiple operating conditions and fault modes) and compare results and calibration across sub-datasets.
- 7. Reporting: document methodology, results, and a discussion of when and how much the model's uncertainty estimates can be trusted.